hold-out task. Therefore, our approach provides a path towards a domain-general model of human cognition, which has been the goal of theoreticians for decades [Newell, 1992, Yang et al., 2019, Riveland and Pouget, 2022, Binz et al., 2023]. We believe that having access to such a model would transform psychology and the behavioral sciences more generally. It could, among other applications, be used to rapidly prototype the outcomes of projected experiments, thereby easing the trial-and-error process of experimental design, or to provide behavioral policy recommendations while avoiding expensive data collection procedures.

Finally, we have to ask ourselves what we can learn about human cognition when finetuning large language models. For now, our insights are limited to the observation that large language model embeddings are rich enough to explain human decision-making. While this is interesting in its own right, it is certainly not the end of the story. Looking beyond the current work, having access to an accurate neural network model of human behavior provides the opportunity to apply a wide range of explainability techniques from the machine learning literature. For instance, we could pick a particular neuron in the embedding and trace back what parts of a particular input sequence excite that neuron using methods such as layer-wise relevance propagation [Bach et al., 2015, Chefer et al., 2021]. Thus, our work also opens up a new spectrum of analyses that are not possible when working with human subjects.

To summarize, large language models are an immensely powerful tool for studying human behavior. We believe that our work has only scratched the surface of this potential and there is certainly much more to come.

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